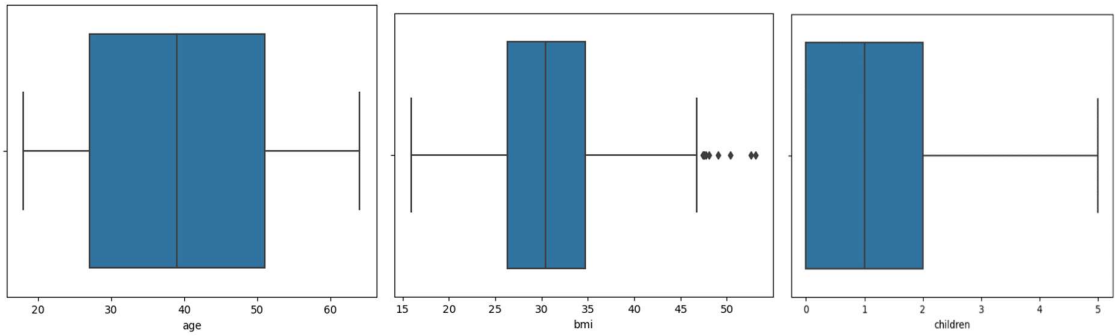
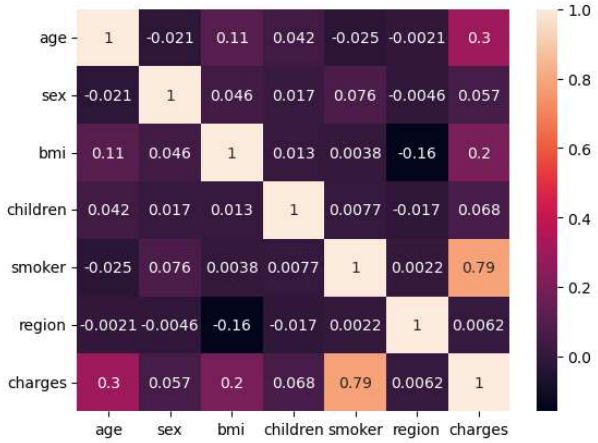
PYTHON FINAL SUBMISSION

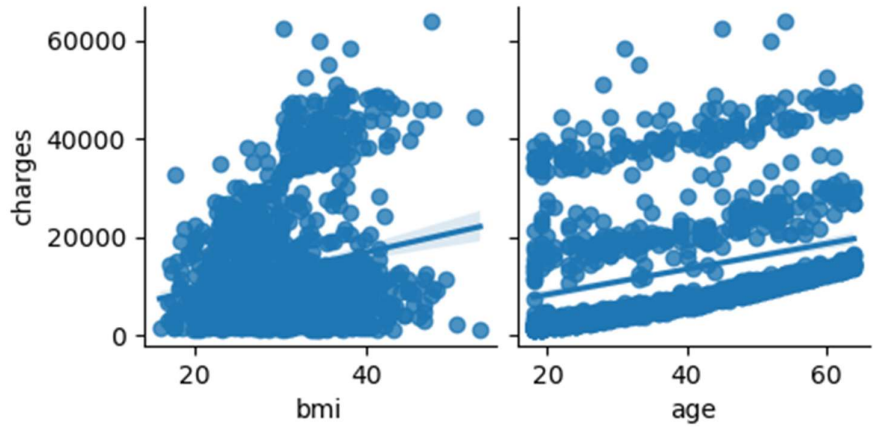
Dataset: Medical Insurance charges forecast



In the first step, we needed to check if there are any outliers in the data. There were outliers in bmi column therefore it needed to be treated.



Correlation Analysis: - There is high correlation found between charges and smoker attributes but since charges is a target variable, we can say they both have significant positive correlation.

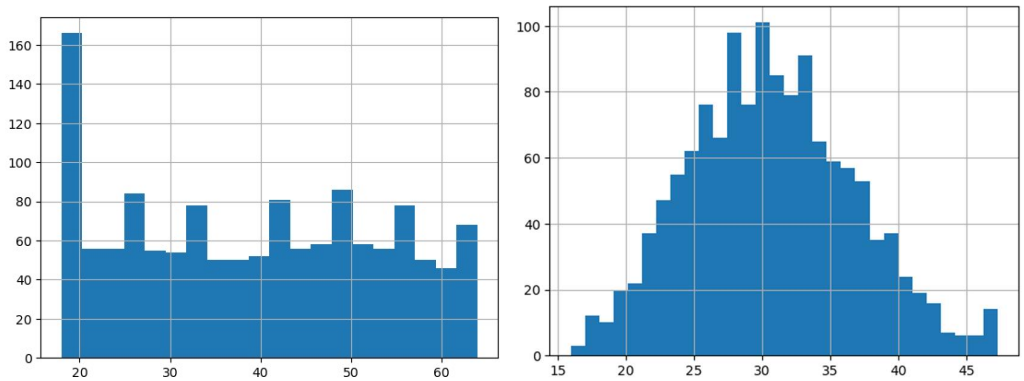


Nothing can be directly understood from the pair plots.

VIF ANALYSIS :

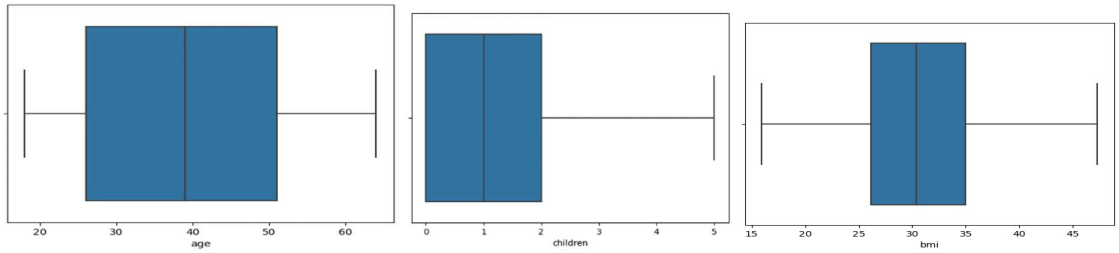
	features	VIF Factor
0	age	8.01
1	sex	8.56
2	bmi	13.35
3	children	1.81
4	smoker	1.26
5	region	5.05

All the independent variables other than BMI are within acceptable range of VIF.



From the above charts, we can see that all data points for bmi attributes are normally distributed but not normally distributed for age column.

After this, we treated the outliers to get better accuracy for the model. From the below diagram, we can see that the outliers are now within the acceptable range.



Splitting the data in Training and Test set

After splitting training data to test data ratio of 75:25 increased accuracy of the model.

Regression results before removing insignificant variables:

OLS Regression Results

Dep. Variable:	charges	**R-squared:**	0.740
Model:	OLS	**Adj. R-squared:**	0.738
Method:	Least Squares	**F-statistic:**	472.5
Date:	Tue, 18 Apr 2023	**Prob (F-statistic):**	2.97e-287
Time:	22:57:10	**Log-Likelihood:**	-10176.
No. Observations:	1003	**AIC:**	2.037e+04
Df Residuals:	996	**BIC:**	2.040e+04
Df Model:	6		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-1.282e+04	1357.119	-9.448	0.000	-1.55e+04	-1.02e+04
age	261.1064	13.996	18.656	0.000	233.642	288.571
sex	25.8395	391.800	0.066	0.947	-743.009	794.688
bmi	308.9572	32.254	9.579	0.000	245.664	372.250
children	460.0915	163.869	2.808	0.005	138.523	781.659
smoker	2.374e+04	492.512	48.201	0.000	2.28e+04	2.47e+04
region	413.9110	181.240	2.284	0.023	58.256	769.566

Omnibus:	239.794	**Durbin-Watson:**	2.029
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	571.165
Skew:	1.276	**Prob(JB):**	9.40e-125
Kurtosis:	5.675	**Cond. No.**	359.

Removed sex attribute from model training due to its insignificance.

Regression results after removing insignificant variables:

OLS Regression Results

Dep. Variable:	charges	**R-squared:**	0.740
Model:	OLS	**Adj. R-squared:**	0.739
Method:	Least Squares	**F-statistic:**	567.6
Date:	Wed, 19 Apr 2023	**Prob (F-statistic):**	1.19e-288
Time:	00:02:36	**Log-Likelihood:**	-10176.
No. Observations:	1003	**AIC:**	2.036e+04
Df Residuals:	997	**BIC:**	2.039e+04
Df Model:	5		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
Intercept	-1.279e+04	1247.722	-10.249	0.000	-1.52e+04	-1.03e+04
age	261.1009	13.988	18.666	0.000	233.651	288.551
bmi	309.0462	32.209	9.595	0.000	245.840	372.252
children	460.4159	163.713	2.812	0.005	139.154	781.678
smoker	2.374e+04	491.493	48.305	0.000	2.28e+04	2.47e+04
region	414.0440	181.138	2.286	0.022	58.589	769.499

Omnibus:	239.755	**Durbin-Watson:**	2.029
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	571.054
Skew:	1.276	**Prob(JB):**	9.94e-125
Kurtosis:	5.675	**Cond. No.**	328.

The above table gives us the following facts:

- The dependent variable is "charges", and there are 5 independent variables

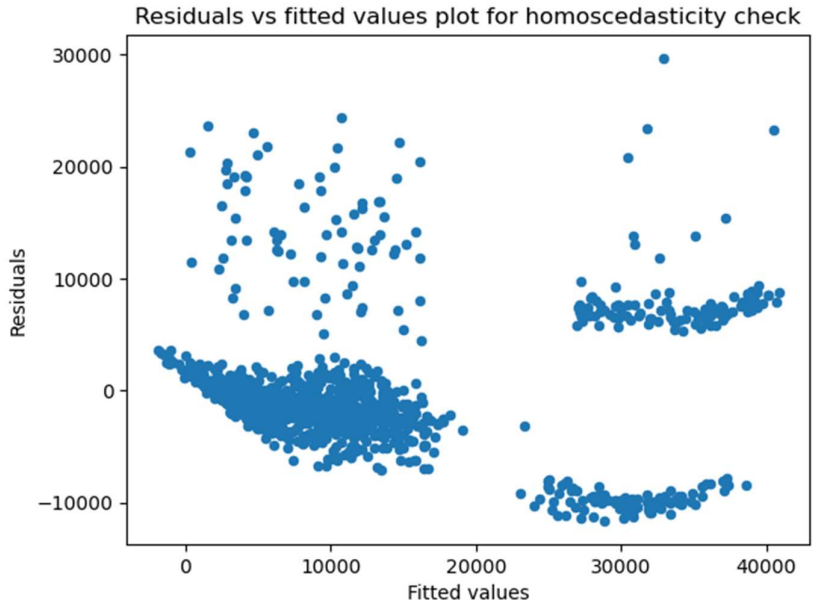
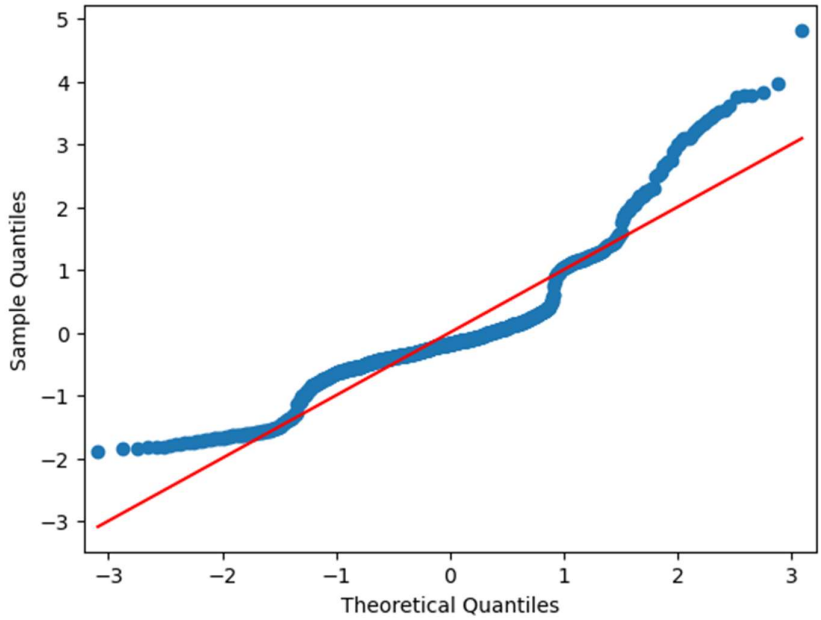
- The model has an R-squared value of 0.740, which means that approximately 74% of the variation in the dependent variable is explained by the independent variables. Also, the adjusted R squared is 0.738 and we can see the difference between both R squared and adjusted R squared is minimal which means the model has better accuracy.

- The F-statistic is **567**.6 which is the critical F value and the Prob F-value is **1.19e-288** which is the calculated F value. The above figures tell us that the overall model is statistically significant and as the gap is high the accuracy of the model is higher.

- The Durbin-Watson tests the model for correlation between the residuals. This value is **2.028616198890516** which suggests that there is no autocorrelation between the errors.

Assumption of Model

Homoscedasticity



Normality